Analytical Chemistry Cumulative Exam
September 10, 2010

Please place all answers in the blue book. For full credit, all work must be shown. Be sure to comment on the logic of your approach. This includes writing out all chemical equations/equilibria.

1) (3 points) Ignoring activity coefficients, the pH of a 1.6 M HNO_3 solution is:_____.

2) (12 points) From Harris, 7th Edition, Exercise 8-A.

Assuming complete dissociation of the salts, calculate the ionic strength of a solution containing 0.2 mM $MgCl_2$ plus 0.3 mM $AlCl_3$.

(The author of this cumulative exam believes you should commit the equation for calculating ionic strength to memory. Thus, the equation is purposely not provided.)

3) (40 points) Ignoring activity coefficients and using mass and charge balance correctly, calculate the pH of a 1.0×10^{-7} M H_2SO_4 solution. Assume at this concentration complete ionization of H_2SO_4 occurs. (As an aside, remember that HSO_4^- is a weak acid with K_a (HSO_4^-) = 1.00×10^{-2}).

4) (50 points)

a) (15 points) Ignoring activity coefficients, calculate the solubility of $Ba(IO_3)_2$ (K_{sp}(standard state conditions) = 1.5×10^{-9}) in water.

b) (25 points) Using the Debye-Hückel Limiting Law, $\log\gamma_i = -0.5z_i^2\sqrt{\mu}$, where γ is the activity coefficient, μ is the ionic strength of the solution, and z is the charge on the ion, calculate the solubility for $Ba(IO_3)_2$ in a solution containing 0.05 M KNO_3.

(If you don't know how to calculate ionic strength, make a guess for the value to solve this problem. Be sure to state you have used a guess for the value.)

c) (10 points) How would you expect the solubility to change between the solution conditions in "a" vs. "b"? Give a chemical explanation for your answer.